DOCUMENT NAME: Meter Interface Control Document
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Revision History

Date. Done
by. The update. New rev.
1 2011/05/12 Carol Primary Edition. Ver1.0
2 2011/07/12 Carol List TD-4222d not supported comm and Ver1.1
3 2011/12/15 Carol Modification Ver1.2
4 2012/4/26 Carol Modification. Ver1.3
5 2013/3/12 Carol Command 0x26 modification. Ver1.4
6 2013/6/3 Carol Command 0x54 modification. Ver1.5
7 2013/09/27 Carol Command 0x26 modification. Ver1.6
8 2017/03/28 Carol Note for different Bluetooth ver sions Ver1.7
9 2018/02/23 Carol Modification. Ver1.8

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1. General

1.1. Overview

The document describes the commands which will be u sed to communicate with blood glucose meter.

Before retrieving data in the device, please make s ure you have already established the proper connection channel between the device and the host side. For example, if you use USB cable, be sure that the USB driver has been properly installe d; or if you use Bluetooth communication, be sure that the connection has been successfully esta blished.

Generally speaking, the device will be the slave si de waiting for the request from the host which is usually the computer , smart phone, or the hub/gate way.

Note for different Bluetooth versions:
For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by for both pairing and connection, the PIN for par ing is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired before any connection, please find the instr uction in meter manual for the pairing process. The meter supports SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:
UUID Base: 1212-efde-1523-785feabcd123
Service UUID: 0x1523
Characteristic: 0x1524 (write/notify)

1.2. Document Overview

For the first time user on the device integration, please read the 2.1 frame structure. If you are familiar with serial port communication, you can re view the command list table in section 2.2 directly. The following sub sections are all about the details for each command.

1.3. Definitions, Acronyms, and Abbreviations

GW Gateway / Hub / Computer side
MD Medica Device, ie. blood glucose meter, blood pr essure meter or scales, etc.
CMD Command. Usually its a one byte alphanumeric s ymbol.
ACK Acknowledgement. Usually its a one byte alphan umeric symbol.
NA Not available

2. Interface specification

Block Diagram

The communication interface is base on RS232 serial port communication :

2.1. Frame Structure

Byte 1 2 3 ~ n-2 n-1 n
Name Start CMD Index, Address, or Data Stop Check
Sum
Format HEX
GW MD 51 CMD A3 [1n-1]
GW MD 51 ACK A5 [1n-1]
ie. Check sum means the summation from byte 1 to b yte n-1 with 8-bit length

Example : 8-byte frame

Byte 1 2 3 4 5 6 7 8
Name Start CMD Data_0 Data_1 Data_2 Data_3 Stop ChkSum
Format HEX
GW MD 51 CMD Data_0 Data_1 Data_2 Data_3 A3 [1..7]
GW MD 51 ACK Data_0 Data_1 Data_2 Data_3 A5 [1..7]
ie. Check sum means the summation from byte 1 to by te 7 with 8-bit length Baud rate 19200
Data bit s 8
Parity No
Start bit 1
Stop bit 1 Medical Device /
Sensor
(MD) Gateway / Hub /
Computer
(GW) TX
RX
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2.2. Command List

Message name CMD/ACK Direction Length
(Bytes) Note
Read device clock time 0x23 GW MD 8
0x23 GW MD 8
Read device model 0x24 GW MD 8
0x24 GW MD 8
Read storage data, part 1(time) 0x25 GW MD 8
0x25 GW MD 8
Read storage data , part 2(result) 0x26 GW MD 8
0x26 GW MD 8
Read device serial number, part 1 0x27 GW MD 8
0x27 GW MD 8
Read device serial number, part 2 0x28 GW MD 8
0x28 GW MD 8
Read storage number of data 0x2B GW MD 8
0x2B GW MD 8
Write device clock time 0x33 GW MD 8
0x33 GW MD 8
Turn off the device 0x50 GW MD 8
0x50 GW MD 8
Clear/Delete all memory 0x52 GW MD 8
0x52 GW MD 8
Notification for entering communication

mo de (only support cable transmission) 0x54 GW MD 8

Note: TD-4222d not support command 0x24 and 0x54

2.2.1. [0x23] Read device clock time (4 bytes)
Message Name: Read device clock time (4 bytes)
Message ID: 0x23
Message Description: Read device date and time (to minute)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time . GW MD 0x23 0x0 0x0 0x0 0x0
Response from device. GW MD 0x23 Day + Month + Year Minute Hour

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x23 GW commands MD to response device clock time
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x23 MD response device clock time
DATA 2 Data_1 + Data_0 : Day+Month+Year (word) (see Table A)
DATA 2 Data_2 : Minute, Data_3 : Hour (see Table B)

Table A: Day + Month + Year

Table B: Minute and Hour MSB LSB
Data_1 Data_0
Year (7-bit) Month (4-bit) Day (5-bit)
6 5 4 3 2 1 0 3 2 1 0 4 3 2 1 0
MSB LSB
Data_ 3 Data_ 2

2.2.2. [0x24] Read device model
Message Name: Read device model
Message ID: 0x24
Message Description: Read device model
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message n ame Direction CMD Data_0 Data_1 Data_2 Data_3
Read device model. GW MD 0x24 0x0 0x0 0x0 0x0

Response from device. GW MD 0x24 device model

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x24 GW commands MD to response device model
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x24 MD response device model
DATA 2 Data_1 + Data_0 : Device model in four digits (Dat a_1 is the higher byte)
DATA 2 Data_3, Data_2 : undefined

2.2.3. [0x25] Read the storage data with index, par t 1(time)
Message Name: Read the storage data with index, par t 1(time) NA Hour (5 -bit) NA Minute (6 -bit)
4 3 2 1 0 5 4 3 2 1 0
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Message ID: 0x25
Message Description: Read data (part 1: measure date and time)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direc tion CMD Data_0 Data_1 Data_2 Data_3
Read the storage data,
part1 . GW MD 0x25 Index 0x0 0x0
Response from device. GW MD 0x25 M_Date M_Time

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x25 GW commands MD to response measure data (part 1)
Index 2 0x00 is the latest reading in the device
DATA 2 0x00 NA

Field Size
(Bytes) Value Description.
ACK 1 0x25 MD response measure date, time and glucose typ e
M_Date 2 The measure date (see Table A in 2.3.1 [0x23])
M_Time 2 The measure time (see Table B in 2.3.1 [0x23])

2.2.4. [0x26] Read the storage data with index, par t 2(result)
Message Name: Read the storage data with index, par t 2(result)
Message ID: 0x26
Message Description: Read data (part 2: measure result)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read the storage data, GW MD 0x2 6 Index 0x0 0x0

part2.
Response from device. GW MD 0x26 Value Type I / Type II

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x26 GW commands MD to response measure data (part 1)
Index 2 0x00 is the latest reading in the device
DATA 2 0x00 NA

Field Size
(Bytes) Value Description.
ACK 1 0x26 MD response measure date and time
Value 2 The measure result (see Table C below)
The unit is mg/dL (mg/dL = mmol/L * 18)

Type I /
Type II 2 Parameters (see Table E below)
[Type I]
0x0: Gen
0x1: AC (before meal)
0x2: PC (after meal)
0x3: QC (quality control)
[Type II]
0x0: Gen.

Table C: Reading
Table D: Parameters MSB LSB
Data_1 Data_0
Value
15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
MSB LSB
Data_3 Data_2
Type I Type II (4 bits) Code (10 bits)
1 0 3 2 1 0 9 8 7 6 5 4 3 2 1 0

2.2.5. [0x27] Read device serial number, part 1
Message Name: Read the device serial number, part 1
Message ID: 0x27
Message Description: Read the device serial number, part 1. This command cooperates with 0x28 to complete the serial number.
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 427211713000001D, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read device serial
number, part 1. GW MD 0x27 0x0 0x0 0x0 0x0
Response from device. GW MD 0x27 SN_0 SN_1 SN_2 SN_3

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x27 GW commands MD to response the device serial
number , part 1
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x27 Read the device serial number, part 1
DATA 4 Data_0 ~ Data_3 : SN_0 ~ SN_3

2.2.6. [0x28] Read device serial number, part 2
Message Name: Read the device serial number, part 1
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Message ID: 0x2 8
Message Description: Read the device serial number, part 2. This command cooperates
with 0x2 7 to complete the serial number.
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 427211713000001D, the former 4 bytes come from 0x28 and the later come from 0x27)

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read device serial
number, part 2. GW MD 0x28 0x0 0x0 0x0 0x0
Response fro m device . GW MD 0x28 SN_4 SN_5 SN_6 SN_7

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x28 GW commands MD to response the device serial
number, part 2
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x28 Read the device serial number, part 2
DATA 4 Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.7. [0x2B] Read storage number of data
Message Name: Read storage number of data
Message ID: 0x2 B
Message Description: Read data counts
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read Storage Number
of Stored Readings GW MD 0x2B 0x0 0x0 0x0 0x0
Response from device GW MD 0x2B Storage Number 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x2B GW commands MD to response the Number of
Stored Readings
DATA 4 0x0

Field Size
(Bytes) Value Description.
ACK 1 0x2B MD response the Number of Stored Readings
DATA 2 Data_1+Data_0 = Storage Number (Word)

2.2.8. [0x33] Write System Clock time (4 bytes)
Message Name: Write device clock time (4 bytes)
Message ID: 0x33
Message Description: Write the device date and time (to minute)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Please refer to 0x23 command section for the format of date and time.

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Write System Clock
time. GW MD 0x33 Day+Month+Year Minute Hour

Response from device. GW MD 0x33 Day+Month+Year Minute Hour

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x33 GW commands MD to write System Clock time (4
bytes)
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_ 2 : Minute, Data_3 : Hour

Field Size
(Bytes) Value Description.
ACK 1 0x33 MD write System Clock time (4 bytes) O.K.
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_2 : Minute, Data_3 : Hour

2.2.9. [0x50] Turn off the device
Message Name: Turn off the device
Message ID: 0x50
Message Description: Turn off the device from G W
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Turn off the device GW MD 0x50 0x0 0x0 0x0 0x0
GW MD 0x50 0x0 0x0 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x50 GW commands MD to turn off the power.

DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x50 MD responses.
DATA 4 0x0 NA

2.2.10. [0x52] Clear/Delete all Memory
Message Name: Clear/Delete all memory
Message ID: 0x52
Message Description: Delete all measurement data in the device
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Clear/Delete all Memory
Storages GW MD 0x52 0x0 0x0 0x0 0x0
Response from device. GW MD 0x52 0x0 0x0 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x52 GW commands MD to Clear/Delete all Memory
Storages in Device EEPROM
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x52 Clear/Delete all Memory Storages in Device EEP ROM
O.K.
DATA 4 0x0 NA

2.2.11. [0x54] Notification for entering communicat ion mode
Message Name: Notification for entering communicati on mode
Message ID: 0x54
Message Description: Some models will send 0x54 command to host as a notification to indicate meter is ready for command from host. Host has to deal with this kind of notification in the beginning of connection (like RS-232) and prevent it to be mixed with the expect command response in receiving queue of host. This command o nly support cable transmission, so bluetooth transmission is no t supported.
Src / Des : MD / GW
Length: Request: 8 bytes / Response: NA

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Notification of entering
communication mode GW MD 0x54 0x0 0x0 0x0 0x0
  No need to response from GW side

Field Description
Field Size
(Bytes) Value Description.
 CMD 1 0x54 MD reports to GW that it enters communication
mode.
 DATA 4 0x0 NA